

WELL COMPLETION REPORT (SYNTHETIC)

OIL-KMC-001 - Kumchai-001

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000012
Field / District	Kumchai / Tinsukia
Block	OALP-IV-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2001-08-19 / 2001-12-07
Measured / TVD	3352 m / 3300 m
Trajectory	Vertical; inc 2.2 deg; az 219.5 deg
Formation Interval	1920-3022 m
Lithology	Interbedded sandstone and shale
Pore Pressure Reference	4135 psi
Fracture Gradient	15.25 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Critical (62.3/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	838	Surface
9 5/8 in casing	1944	Intermediate
7 in liner/casing	2949	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2001-09-22 1203 m	Kick	Critical	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.
2001-11-18 2827 m	Stuck Pipe	Critical	Torque and drag increased while ROP declined. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2001-09-02 575 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Avoid prolonged static periods in high-drag interval.
2001-09-10 828 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2001-10-24 2118 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2001-11-08 2552 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2001-11-11 2638 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 3352 m MD in the Barail interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.